

Verticus Farms — vertical farming for leafy greens and herbs

The Problem

Leafy greens travel an average of 1,900 kilometres to reach a European supermarket, arrive with a shelf life of four days, and are grown in regions where water is becoming the binding constraint. Around 30% of the crop is lost between field and shelf.

Growing them beside the city that eats them removes the distance, the loss and the water.

Our Solution

Verticus operates indoor vertical farms producing lettuce, basil, coriander and rocket, in stacked growing trays under tuned LED lighting with a recirculating hydroponic system.

Our facility outside Lyon has been in continuous production for two years. Yield is 94 kilograms per square metre per year, against roughly 4 in open field. Water use is 3% of field equivalent. There are no pesticides, harvest is twelve days after seeding, and the product reaches the shelf the same day it is cut with a shelf life of eleven days.

The growing system works. We have run it for two years and the agronomy is not the uncertainty here.

The Market

The European market for leafy greens and fresh herbs is approximately €9B at retail.

Our production cost places us above field-grown produce and below most imported organic, which positions us in the premium segment of that market rather than across it.

Competition

Several vertical farming companies operate in Europe. Three large ones have closed or restructured in the last two years, all of them for the same reason: energy cost against a commodity selling price.

We are not going to claim we have solved what they did not.

The Team

Élodie Pichon has a doctorate in plant science and led the agronomy programme at a horticultural research institute. Marc Dietrich ran production at a food manufacturer for eight years.

Twenty-six people.

Traction

- facility in continuous production for two years
- 94 kg per square metre per year sustained yield
- supply agreements with two regional supermarket chains and eleven restaurants
- 340 tonnes produced last year
- product quality consistently rated above the imported comparison in the buyers' own testing

Business Model

Here is the problem, stated directly.

Our all-in production cost is €4.10 per kilogram. Energy is €2.30 of that, and it is the line that has moved against us — it was €1.40 when the facility was designed.

Field-grown lettuce delivered to a French supermarket distribution centre costs between €1.10 and €1.80 per kilogram. Imported organic runs €2.60 to €3.40.

We sell at €3.90. We lose money on every kilogram we ship, and the two supermarket chains that buy from us have both refused a price increase, correctly, because they can buy the same product for less.

We have taken €0.60 per kilogram out of the cost base in two years, mostly through lighting efficiency. We need to take out another €1.50 and we do not currently know where it comes from.

The customer cannot pay what it costs us to grow. That is the whole business in one sentence.

Funding

We are raising €10M to fund eighteen months of operation while we attempt to close the cost gap.